Enrollment No.: _____

Darshan UNIVERSITY

Darshan Institute of Engineering & Technology
B.Tech. | Sem-4 | Summer-2025

Course Code	:	2301CS401	**Date**	:	10-04-2025
Course Name	:	Database Management System - II	**Duration**	:	150 Minutes
			Total Marks	:	70

Instructions:
1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

Q.1 **(A)** What are the key differences between a Stored Procedure and a Function in SQL? **4**

(B) What is a Cursor in SQL? Explain its types and provide an example where it is required. **3**

OR

What is a Cursor in SQL? Explain its life cycle.

(C) Employees (EmployeeID, EmployeeName, DepartmentID) **7**
Departments (DepartmentID, DepartmentName)
Create a stored procedure that accepts a department name as input and returns the total number of employees in that department.

OR

Employees (EmployeeID, EmployeeName, DepartmentID)
Departments (DepartmentID, DepartmentName)
Create a function that calculates the total number of employees in each department. (Consider column names as per your choice.)

Q.2 **(A)** What is transaction? Explain ACID property. **4**

(B) Explain States Transitions Diagram with the help of diagram. **3**

OR

Explain modes of transactions in SQL server.

(C) Employees(EmpID, EmpName, Salary, DepartmentID) **7**
History(Description)
Create a trigger that insert a message 'An employee whose ID is XXX salary is updated and new salary is Rs. XXX' as new record in the History table whenever an employee's salary is updated.

OR

Employees(EmpID, EmpName, Salary, DepartmentID)
Departments(DepartmentID, DepartmentName, TotalSalary)
Create a trigger that updates the TotalSalary column in the Departments table whenever an employee's salary is updated.

Q.3 **(A)** Differentiate Clustered Index and Non-Clustered Index. **4**

(B) Explain types of execution plan. **3**

OR

Explain types of temporary table.

(C) What are the common warnings found in an execution plan? **7**

OR

What are the best practices for writing efficient SQL queries?

Q.4 **(A)**

1) What is the Full form of JSON? **4**
2) Which command is used to show the list of Databases.
3) What does first 4 bytes of objecid represents in mongodb?
4) What is the size of objectid in mongodb?

(B) Explain different types of NoSQL database. **3**

OR

Explain disadvantages of NoSQL database.

(C) **7**

{_id : 1, Name : {Fname: "Neha", Lname: "Sharma"}, Dept: ‘CE’, SPI : 9.0, BL : 0 },
{_id : 2, Name : { Fname:"Milan", Lname: "Patel"}, Dept: ‘CE’, SPI : 9.5, BL : 0 },
{_id : 3, Name : { Fname:"Sohan", Lname: "Dave"}, Dept: ‘IT’, SPI : 8.6, BL : 1 },
{_id : 4, Name : { Fname:"Nehal", Lname: "Patel"}, Dept: ‘IT’, SPI : 7.0, BL : 3 },
{_id : 5, Name : { Fname:"Milani", Lname: "Dave"}, Dept: ‘CE’, SPI : 5.5, BL : 5 },
{_id : 6, Name : { Fname:"Jay", Lname: "Sharma"}, Dept: ‘EC’, SPI : 8.0, BL : 2 }

1. Display all documents where the department is either "CE" or “IT” and SPI is greater than or equal to 8.0.
2. Update the last name (Lname) of the document where _id is 2, set it to "Mehta".
3. Increase the backlog (BL) by 1 for students belonging to the "IT" department who have an SPI of less than 8.0.
4. If a document with _id 7 exists, update its SPI to 9.8. If no such document exists, insert a new document with _id 7 and set its SPI to 8.0.

OR

{_id : 1, Name : {Fname: "Neha", Lname: "Sharma"}, Dept: ‘CE’, SPI : 9.0, BL : 0 },
{_id : 2, Name : { Fname:"Milan", Lname: "Patel"}, Dept: ‘CE’, SPI : 9.5, BL : 0 },
{_id : 3, Name : { Fname:"Sohan", Lname: "Dave"}, Dept: ‘IT’, SPI : 8.6, BL : 1 },
{_id : 4, Name : { Fname:"Nehal", Lname: "Patel"}, Dept: ‘IT’, SPI : 7.0, BL : 3 },
{_id : 5, Name : { Fname:"Milani", Lname: "Dave"}, Dept: ‘CE’, SPI : 5.5, BL : 5 },
{_id : 6, Name : { Fname:"Jay", Lname: "Sharma"}, Dept: ‘EC’, SPI : 8.0, BL : 2 }

1. Display all documents where the department is either "CE" or “IT” and SPI is between 8.0 and 9.0.
2. Insert a new field named JoiningDate with value as todays date in every document.
3. Display Fname, Dept and SPI of student belongs to CE or IT department whose SPI is between 8 and 9.
4. Decrement the backlog (BL) by 1 for students of "IT" department having SPI less than 8.0.

Q.5 **(A)** Define Replication and why it is required. **4**

(B) Explain BASE Theorem. **3**

OR

Explain CAP Theorem.

(C) { "_id": 1,"username": "john_doe", "email": "john.doe@gmail.com", "phone": "9876543210", "customerName": "Johnathan Doe" }, **7**
{ "_id": 2, "username": "macUser123", "email": "mac.user@icloud.com", "phone": "7894561230", "customerName": "Mac Johnson" },
{ "_id": 3, "username": "noNumbersHere", "email": "unique.email@xyz.com", "phone": "1231231234", "customerName": "Alice Wonderland" },
{ "_id": 4, "username": "sp3cial!user", "email": "special@domain.com", "phone": "9876512345", "customerName": "Special User" },
{ "_id": 5, "username": "shorty", "email": "short.short@short.com", "phone": "9988776655", "customerName": "Sally Five" }

1. Display name of users whose email belongs to gmail.com.
2. Find Products with Names Starting with 'Mac'.
3. Find Products with Names End with 'Pro' or 'Plus'.
4. Search for users having a 10-Digit Phone Number.

OR

{ "_id": 1, "customer": "Alice", "totalAmount": 150, "category": "Electronics", "date": "2024-03-01" },
{ "_id": 2, "customer": "Bob", "totalAmount": 200, "category": "Clothing", "date": "2024-03-02" },
{ "_id": 3, "customer": "Alice", "totalAmount": 300, "category": "Electronics", "date": "2024-03-05" },
{ "_id": 4, "customer": "Charlie", "totalAmount": 400, "category": "Furniture", "date": "2024-03-10" },
{ "_id": 5, "customer": "Alice", "totalAmount": 100, "category": "Clothing", "date": "2024-03-12" },
{ "_id": 6, "customer": "Bob", "totalAmount": 250, "category": "Electronics", "date": "2024-03-15" },
{ "_id": 7, "customer": "Charlie", "totalAmount": 350, "category": "Furniture", "date": "2024-03-20" }

1. Find the number of orders placed by each customer.
2. Find the Maximum, Minimum and Total Order Amount Per Category.
3. Get the earliest order date for each customer.
4. Display category-wise total amount which do not contains Clothing category and sort the output on category in descending order.

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